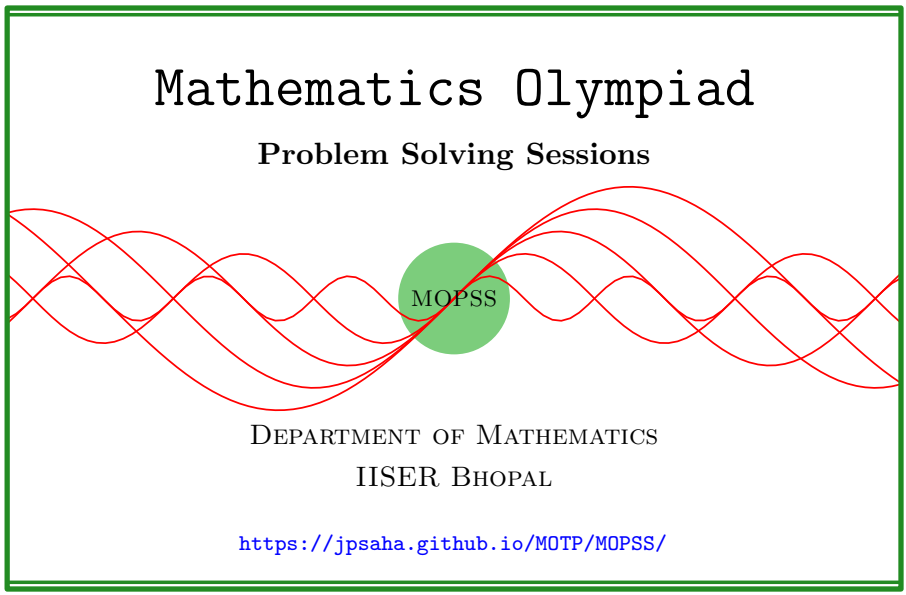
Orders

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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§1 Order

Definition 1 (Order of an integer modulo a prime)**.** Let p be a prime, and a be an integer, not divisible by p. The **order of a modulo p**, denoted by $\mathrm{ord}_p(a)$, is defined to be the smallest positive integer such that $a^{\mathrm{ord}_p(a)} \equiv 1 \bmod p$.

Lemma 2

Let p be a prime, and a be an integer, not divisible by p. Let k be an integer such that $a^k \equiv 1 \bmod p$. Then, $\mathrm{ord}_p(a)$ divides k. In particular, $\mathrm{ord}_p(a)$ divides $p-1$. Moreover, if $a^k \equiv 1 \bmod p$, then $a^{\gcd(k,\mathrm{ord}_p(a))} \equiv 1 \bmod p$.

Exercise 1. For any prime p, any prime divisor q of 2^p+1 satisfies $q = 3$ or $2p$ divides $q-1$.

Proof. Note that the order of 2 modulo q divides $2p$. Since the order of 2 modulo q cannot be 1 or p, it follows that the order of 2 modulo q is equal to 2 or $2p$. If the order of 2 modulo q is equal to 2, then $q = 3$. If the order of 2 modulo q is equal to $2p$, then $2p$ divides $q-1$, and hence, $q > 2p$. □

Exercise 2 (RMO 2014a P3, cf. Tournament of Towns, Fall 1988, Training)**.** [Tao06, Problem 2.2] [AE11, Problem 3.81] Suppose for some positive integers r and s, the integer 2^r is obtained by permuting the digits of 2^s in decimal expansion and $2^r, 2^s$ have same number of digits. Prove that $r = s$.

Walkthrough —

(a) On the contrary, assume that $r \neq s$. Without loss of generality, we may assume that $r > s$ holds.

(b) Show that $2^{r-s} - 1$ is divisible by 9.

The content posted here and at this blog by Evan Chen are quite useful.

(c) Prove that 6 divides $r - s$.

(d) Conclude that $2^r \geq 64 \cdot 2^s$ holds, contradicting the fact that 2^r and 2^s have the same number of digits.

Solution 1. Since a positive integer is congruent to the sum of its digits modulo 9, it follows that 2^r and 2^s are congruent modulo 9.

Let us consider the case that $r < s$. Note that 9 divides $2^{s-r} - 1$. Since the order of 2 modulo 9 is equal to 6, it follows that 6 divides $s - r$, and hence $2^s \geq 64 \cdot 2^r$, which is impossible. This shows that $r \geq s$ holds. Similarly, it also follows that $s \geq r$ holds. This proves that $s = r$, as required. ■

Exercise 3 (IMO Shortlist 2010 N2, , proposed by Angelo Di Pasquale, Australia, cf. India IMOTC 2011, Mathematical Ashes 2011 P2)**.** Find all pairs (m, n) of non-negative integers for which

$$m^2 + 2 \cdot 3^n = m(2^{n+1} - 1).$$

Walkthrough —

(a) Let m, n be nonnegative integers satisfying the given equation. Considering the roots of $x^2 - x(2^{n+1} - 1) + 2 \cdot 3^n$, it follows that

$$3^k + 2 \cdot 3^\ell = 2^{n+1} - 1$$

holds, for some nonnegative integers k, ℓ satisfying $k + \ell = n$.

(b) Note that

$$3^k < 2^{n+1} < 9^{(n+1)/3}$$

holds, implying $k < 2(n+1)/3$. Also note that

$$2 \cdot 3^\ell < 2^{n+1} < 2 \cdot 3^{2n/3}$$

holds, implying $\ell < 2n/3$. Using $k + \ell = n$, show that

$$k > \frac{n-2}{3}, \ell > \frac{n-2}{3}.$$

(c) Let us consider the case[a] that $n \geq 6$. Note that $m := \min\{k, \ell\} \geq 2$ holds.

(i) Note that 9 divides $2^{n+1} - 1$, and show that 6 divides $n + 1$. Writing $n + 1 = 6j$ yields

$$2^{n+1} - 1 = (4^j - 1)(4^{2j} + 4^j + 1) = (2^j - 1)(2^j + 1)((4^j - 1)^2 + 3 \cdot 4^j).$$

(ii) Noting that $(4^j - 1)^2 + 3 \cdot 4^j$ is divisible by 3, but not by 9, and that the integers $2^j - 1, 2^j + 1$ are coprime, conclude that 3^{m-1} divides one of $2^j - 1, 2^j + 1$.

(iii) Prove that

$$3^{m-1} \leq 2^j + 1 \leq 3^j = 3^{\frac{n+1}{6}},$$

implying

$$m - 1 \leq \frac{n+1}{6}.$$

(iv) Conclude that

$$\frac{n-2}{3} - 1 < m - 1 \leq \frac{n+1}{6}.$$

holds.

(v) This yields $n < 11$, contradicting $n \geq 6$ and 6 divides $n + 1$.

(d) It remains to consider the case $n \leq 5$.

[a]It also suffices to assume that $n \geq 5$ holds to obtain $m \geq 2$.

Solution 2. Let m, n be non-negative integers satisfying satisfying the given equation. Since the quadratic polynomial $x^2 - x(2^{n+1} - 1) + 2 \cdot 3^n$ has one integer root, it follows that its roots are integers, and are equal to $3^k, 2 \cdot 3^\ell$ for some nonnegative integers k, ℓ satisfying $k + \ell = n$. This gives

$$3^k + 2 \cdot 3^\ell = 2^{n+1} - 1.$$

Using $2^{n+1} < 3^{\frac{2(n+1)}{3}}$, we have

$$3^k < 3^{\frac{2(n+1)}{3}}$$

and

$$2 \cdot 3^\ell < 2^{n+1} < 2 \cdot 3^{\frac{2(n+1)}{3}}$$

hold, which shows that

$$k < \frac{2(n+1)}{3}, \ell < \frac{2(n+1)}{3}.$$

Since $k + \ell = n$, we obtain

$$k > \frac{n-2}{3}, \ell > \frac{n-2}{3}.$$

Let us consider the case that $n \geq 5$. Let m denote the minimum of the integers k, ℓ. Note that $m \geq 2$ holds. Since 9 divides $2^{n+1} - 1$, it follows that 6 divides $n + 1$. Write $n + 1 = 6j$ for some positive integer j. This gives

$$2^{n+1} - 1 = (4^j - 1)(4^{2j} + 4^j + 1) = (2^j - 1)(2^j + 1)((4^j - 1)^2 + 3 \cdot 4^j).$$

Note that $(4^j - 1)^2 + 3 \cdot 4^j$ is divisible by 3, but not by 9, and that the integers $2^j - 1, 2^j + 1$ are coprime being consecutive odd integers. This shows that 3^{m-1} divides one of $2^j - 1, 2^j + 1$. It follows that

$$3^{m-1} \leq 2^j + 1 \leq 3^j = 3^{\frac{n+1}{6}},$$

which yields

$$m - 1 \leq \frac{n+1}{6}.$$

Combining this with

$$\frac{n-2}{3} < m,$$

we obtain

$$\frac{n-2}{3} - 1 < \frac{n+1}{6},$$

which gives $n < 11$. Since $n+1$ is divisible by 6 and $n \geq 5$, it follows that $n = 5$ holds. Therefore, it remains to consider the case that $n \leq 5$.

Note that the discriminant of the quadratic polynomial $x^2 - x(2^{n+1}-1) + 2 \cdot 3^n$ is equal to

$$(2^{n+1} - 1)^2 - 4 \cdot 2 \cdot 3^n,$$

which is a perfect square. This implies that n is not equal to any of $0, 1, 2, 4$. Consequently, for any nonnegative integers m, n satisfying the given equation, the integer n is equal to either 3 or 5. Checking these cases, we obtain the solutions

$$(6, 3), (9, 3), (9, 5), (54, 5),$$

which are precisely all the solutions of the given equation over the nonnegative integers. ■

Exercise 4 (Dutch BxMO/EGMO TST 2025 P5, cf. Korea Final Round 2023 P4,). Determine all positive integers n for which all prime factors of $2^n - 1$ are at most 7.

> **Walkthrough** —
>
> **(a)**

Exercise 5 (Germany TST 2010 Day 1 P3, ,). Determine all (m, n) of positive integers satisfying $3^m - 7^n = 2$.

Exercise 6 (Romania JBMO TST 2025 Day 2 P4, ,). Determine all natural numbers n such that $2^n - n^2 + 1$ is a perfect square.

> **Walkthrough** —
>
> **(a)** Verify that $n = 2$, 3 and 4 works.
>
> **(b)** Show that if $n \geq 5$ is even, then $2^n - n^2 + 1$ is not a perfect square.
>
> **(c)** If $n \geq 5$ is odd, then consider the largest prime divisor p of n.
>
> **(d)** If 9 divides $2^p + 1$, then determine p modulo 6, and show that 19 divides $2^n + 1$, and obtain a contradiction.
>
> **(e)** If 9 does not divide $2^p + 1$, then show that $2^p + 1$ admits a prime divisor

$q > p$ such that q is congruent to 3 modulo 4, and obtain a contradiction.

Solution 3. Note that $2^n - n^2 + 1$ is a perfect square if n is equal to 2, 3 or 4. Let us now show that there are no other values of n for which $2^n - n^2 + 1$ is a perfect square. Let $n \geq 5$ be a positive integer such that $2^n - n^2 + 1$ is a perfect square. Let us first consider the case when n is even. Let $n = 2k$ for some positive integer k. Then, we have

$$2^n - n^2 + 1 = 2^{2k} - 4k^2 + 1 < (2^k)^2,$$

which implies that

$$2^n - n^2 + 1 \leq (2^k - 1)^2.$$

It follows that $2^k \leq 2k^2$, and thus $k \leq 6$. If n is divisible by 6, then $2^n - n^2 + 1$ is congruent to 2 modulo 3, and thus it is not a perfect square. Hence, n is equal to 8 or 10. If n is equal to 8, then $2^n - n^2 + 1$ is not a perfect square. If n is equal to 10, then $2^n - n^2 + 1$ is not a perfect square. This proves that there are no even values of n greater than 4 such that $2^n - n^2 + 1$ is a perfect square.

Next, let us consider the case when n is odd. Let a be a nonnegative integer such that $2^n + 1 = n^2 + a^2$. Let p be the largest prime divisor of n. Since n is odd, it follows that $2^p + 1$ divides $2^n + 1$. Note that $2^p + 1$ is divisible by 3. Let us consider the case when 9 divides $2^p + 1$. Note that $p - 3$ is divisible by 6, and thus $p = 3$. Since $n \geq 5$ and p is the largest prime divisor of n, it follows that 9 divides n, and hence, $2^9 + 1$ divides $2^n + 1$. Since $2^9 + 1$ is divisible by 19, we obtain that 19 divides $2^n + 1$. This implies that 19 divides $n^2 + a^2$. Since 19 is congruent to 3 modulo 4, we have that -1 is not a quadratic residue modulo 19. It follows that 19 divides both n and a. This contradicts the fact that $p = 3$ is the largest prime divisor of n. Hence, 9 does not divide $2^p + 1$.

Since $2^p + 1$ is congruent to 1 modulo 4, and it is divisible by 3, but not by 9, we conclude that $2^p + 1$ admits a prime divisor $q > 3$ such that q is congruent to 3 modulo 4. This implies that q divides $n^2 + a^2$. Since q is congruent to 3 modulo 4, we have that -1 is not a quadratic residue modulo q. It follows that q divides both n and a. By the following claim, we have that $q > p$, which contradicts the fact that p is the largest prime divisor of n.

Claim — For any prime p, any prime divisor q of $2^p + 1$ satisfies $q = 3$ or $2p$ divides $q - 1$.

Proof of the claim. Note that the order of 2 modulo q divides $2p$. Since the order of 2 modulo q cannot be 1 or p, it follows that the order of 2 modulo q is equal to 2 or $2p$. If the order of 2 modulo q is equal to 2, then $q = 3$. If the order of 2 modulo q is equal to $2p$, then $2p$ divides $q - 1$, and hence, $q > 2p$. □

■

References

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[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)

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